MIL-T-58093A(AV)
5 January 1976
Superseding
MIL-T-58093
24 May 1971

MILITARY SPECIFICATION

TESTER, GYRO INDICATOR, PITCH AND ROLL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a gyro indicator pitch and roll tester.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-575	Bolt, Hexagon and Square
FF-N-836	Nut: Square, Hexagon, Cap, Slotted, Castellated, Clinch, Knurled, Welding
FF-S-85	Screw, Cap, Slotted- and Hexagon-Head
FF-S-92	Screws, Machine: Slotted, Cross-Recessed

FSC 4920

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FF-S-107	Screws, Tapping and Drive
FF-W-84	Washers, Lock (Spring)
FF-W-92	Washers, Metal, Flat (Plain)
QQ-A-250/5	Aluminum Alloy Alclad -- 2024, Plate and Sheet
QQ-B-613	Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet and Strip)
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-M-31	Magnesium Alloy, Bars, and Special Shaped Sections, Extruded
QQ-M-40	Magnesium Alloy Forgings
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-B-601	Box, Wood, Cleated-Plywood

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MIL-P-116	Preservation, Packaging, Methods of
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-L-2105	Lubricating Oil, Gear, Multi-purpose
MIL-M-3171	Magnesium Alloy, Processes for Pre-treatment and Prevention of Corrosion on
MIL-C-5015	Connectors, Electric, AN Type
MIL-I-5133	Indicator, Attitude, Type J-8
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-R-5674	Rivet, Aluminum and Aluminum Alloy
MIL-I-5741	Indicator, Directional, Gyroscopic, Type C-5C
MIL-L-6086	Lubricating Oil, Gear, Petroleum Base
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals

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MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-T-21200	Test Equipment for Use With Electrical Equipment, General Specification for

STANDARDS

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Markings of US Military Property
MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-810	Environmental Test Methods
MS3102	Connector, Receptacle, Electric, Box Mounting
MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form of

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.--The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

NATIONAL FIRE PROTECTION ASSOCIATION

Pamphlet No. 70 National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

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Processes	Specification
Cadmium	QQ-P-416
Zinc	QQ-Z-325
Nickel	QQ-N-290
Chromium	QQ-C-320
Anodizing of aluminum	MIL-A-8625
Chemical treatment of aluminum	MIL-C-5541
Protective treatment of magnesium	MIL-M-3171
Oxide coating	MIL-C-13924

3.3.3 Soldering.--Soldering shall be in conformance with MIL-S-6872.

3.3.4 Threaded parts.--Screw threads 0.060-inch or larger in diameter shall be in accordance with MIL-S-7742. Screw threads under 0.060-inch in diameter shall be in conformance with Handbook H28.

3.3.5 Standard parts.--AN or MS standard parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawings by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used provided they have suitable properties and are capable of being replaced by AN or MS parts without alteration. In application for which no suitable corresponding AN or MS part is in effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

3.3.6 Interchangeability of parts.--All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.3.7 Maintainability.--Tester design shall be such that disassembly, reassembly, and servicing of any unit, can be done with standard tools and equipment, with minimum disassembly of other components.

3.4 Major components.--All components shall be of nonmagnetic metal. The metal shall be suitably protective to prevent corrosion.

3.4.1 Base.--The tester base shall be constructed so that all controls are housed in the base.

3.4.2 Cradle.--The cradle shall be mounted to the base, and arranged for positioning either horizontal or vertical. The cradle design shall permit mounting of a reversible drive motor with a reduction gear head which will rotate the fork in either direction.

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3.4.3 Fork and gimbal ring.--The fork design shall provide for mounting the gimbal ring so that when the fork is in operation it will be driven through a clutch by the shaft of the motor reduction gear head, and rotated in a horizontal or vertical plane according to the position of the cradle.

3.4.4 Mounting plates.--The mounting plate for assembly within the gimbal ring shall be designed and constructed for mounting four support posts, to permit the testing of larger type indicators than specified in table I. A mounting plate shall also be provided to test the type J-8 attitude gyro indicator in accordance with MIL-I-5133. Another plate shall also be provided for testing the C-5C directional gyro indicator in accordance with MIL-I-5741.

3.4.5 Adapter plates.--Adapter plates shall be provided for mounting to the mounting plates, in accordance with table I.

3.4.6 Circular levels.--One circular level shall be applied to the gimbal ring, and one onto the fork.

3.4.7 Drive motor.--The drive motor shall be a hysteresis type, synchronous, with an adjustable resistor to permit resetting the angular motor speed. The drive motor shall comply with the National Electrical Code for hazardous locations, Article 500, Class 1, Division 2 of NFPA Pamphlet No. 70.

3.4.8 Vibrator motor.--The vibrator motor shall have a speed of 1,550 revolutions per minute (rpm), with vibration settings of 1,550 cycles with amplitude from 0.002- to 0.005-inch. The vibrator motor shall comply with the National Electrical Code for hazardous locations, Article 500, Class 1, Division 2 in accordance with NFPA Pamphlet No. 70.

3.4.9 Reduction gear head.--The reduction gear head, for use in conjunction with the drive motor, shall be mounted on the cradle. The reduction gear head shall have a filler plug located on top of the housing, for servicing the gear housing oil for the temperature ranges listed in table II:

TABLE II. -- Lubricants for temperature ranges.

Temperature range		Lubricant	
Centigrade	Fahrenheit	Specification	Commercial designation
Minus 46° to 4.5°	Minus 50° to 40°	MIL-L-6086	Mobile gear oil medium (no substitute)
0° to 32°	32° to 90°	MIL-L-2105	Metalicoil A-125 Stano gear leaded lubricant 3X
32° to 52°	90° to 125°	MIL-L-2105	No. 220 gear oil

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3.4.10 Controls.--The controls shall be housed in the base (see 3.4.1). A speed selector switch shall be provided to select factory set speeds of OFF, 10, 15, and 20 degrees per second. The maximum deviation of the turn rate shall not exceed plus or minus 2 percent. One toggle switch shall be provided to turn the vibration motor ON or OFF, and another for selecting the direction of rotation of the three-speed drive motor. Two indicator lights and two fuses shall indicate the functioning of the motors and protect the supply circuits.

3.5 Electrical equipment.--Electrical parts and their application shall be in accordance with MIL-E-5400. Parts that do not appear on the approved lists in accordance with MIL-E-5400 shall not be used without prior written approval by the procuring activity.

3.5.1 Wiring.--Wiring shall be cable-laced or adequately secured. All wiring shall have lugs. All splices shall be made with approved connectors. An equipment ground terminal lug shall be attached securely to the tester. Wiring conductors shall be in accordance with MIL-W-16878.

3.5.2 Receptacles.--Electrical receptacles shall be in accordance with MIL-C-5015.

3.5.3 Power source.--Power shall be from an external source, 117 volts, plus or minus 5 percent, 60 cycles, single phase.

3.5.4 Fuses.--Fuses shall be in accordance with MIL-T-21200.

3.5.5 Under-voltage protection.--The tester shall not be damaged by operating on power line voltage to below the minimum specified in short periods (3.5.3), and shall automatically resume normal operation when the voltage returns within limits.

3.5.6 Power connectors.--The tester shall enable the input power cables to enter through the rear of the machine. The connectors shall be in accordance with MS3102. Connecting cables 9 feet long shall be provided.

3.5.7 Power switch.--A power selector switch shall control application of power to the tester. The switch shall be mounted on the front of the tester base. A pilot lamp shall be mounted to indicate power ON.

3.6 Performance.--The tester shall perform satisfactorily when it meets all tests specified in section 4.

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3.7 Environmental requirements.--The tester shall be designed to operate in the following environmental conditions of MIL-STD-810.

Requirement	Method	Procedure
High Temperature	501	I (Highest operating temperature 52°C (125°F))
Low temperature	502	I (Lowest operating temperature Minus 32°C (minus 25°F))

3.8 Radio-interference suppression.--All radio-interference producing equipment installed within the tester shall be suppressed in accordance with 4.4.5.

3.9 Marking of assemblies.--All markings shall be durable to withstand usage encountered in service. Letters and numerals shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.10 Identification of product.--Tester shall be identified in accordance with MIL-STD-130. The nameplate of the tester, and all subassemblies, shall be in accordance with MIL-P-514.

3.11 Workmanship.--The tester, including all parts and accessories, shall be constructed and finished to produce a product free from defects which could affect proper functioning in service. Particular attention shall be given to neatness, thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding, brazing, painting, machine screw assemblies, and freedom of all parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of examinations and tests.--Examinations and tests of the tester shall be classified as (1) first article, (2) quality conformance.

4.2.1 First article.--The first tester (see 3.1.1) shall be submitted to all tests specified in 4.3 and 4.4.

4.2.2 Quality conformance.--Each tester shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3.

4.3 Test conditions.--

4.3.1 Attitudes.--Unless otherwise specified, the tester shall be tested in the normal operating position.

4.3.2 Tapping.--Unless otherwise specified, the tester indicators and controls may be gently tapped or vibrated while test readings are taken.

4.4 Test procedures.--

4.4.1 Examination of product.--A visual inspection shall be made to verify that the materials, design and construction, mechanical measurements, marking, and workmanship comply with this specification.

4.4.2 Electrical.--A continuity check shall be made on all accessory cables and motors.

4.4.3 Operation.--The tester shall be operated according to the manufacturer's instructions. Tester shall be tested to determine conformance with 3.4.8 and 3.4.10. The test results shall be used for pretest data of the environmental test.

4.4.4 Environmental.--Environmental tests shall be in accordance with the following methods and procedures specified in MIL-STD-810:

Tests	Method	Procedure
High temperature	501	I (Highest operating temperature 52°C (125°F))
Low temperature	502	I (Lowest operating temperature minus 32°C (minus 25°F))

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4.4.5 Radio-interference suppression.--The tester shall be tested for conformance with MIL-STD-461.

4.5 Inspection of preparation for delivery.--Preservation and packaging, packing, and marking shall be inspected to determine conformance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.--Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.--Each tester shall be packaged method IIa of MIL-P-116 in a unit container conforming to Grade A of PPP-B-601. The extension cables shall be coiled, and the cables and mounting plates shall be secured in a manner to prevent movement within the package.

5.1.2 Level C.--Each tester shall be packaged in a manner to prevent deterioration and damage during handling and shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing.--Testers packaged per 5.1.1 shall require no packing. Testers packaged per 5.1.2 shall be packed in a manner to assure carrier acceptance and safe delivery to destination.

5.3 In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.--The tester is intended for use in testing the performance characteristics of various aircraft instrument components such as directional heading indicators, attitude horizon indicators, and directional gyro transmitters. Representative types of aircraft instrument components which will be tested should be governed by the applicable specifications.

6.2 Ordering data.--Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether first article approval is required (see 3.1.1 and 4.2.1).

(c) Level of preservation and packaging, and packing required (see 5.1 and 5.2).

(d) Whether special marking for shipment is required (see 5.3).

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Preparing Activity:

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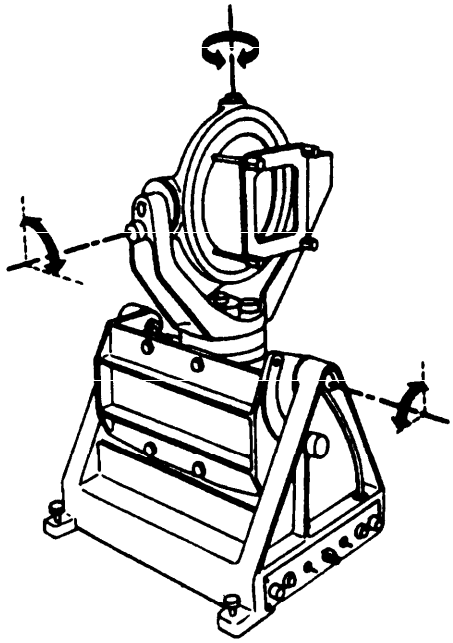


Figure 1(a). Tester with fork and cradle in turn (vertical) position.

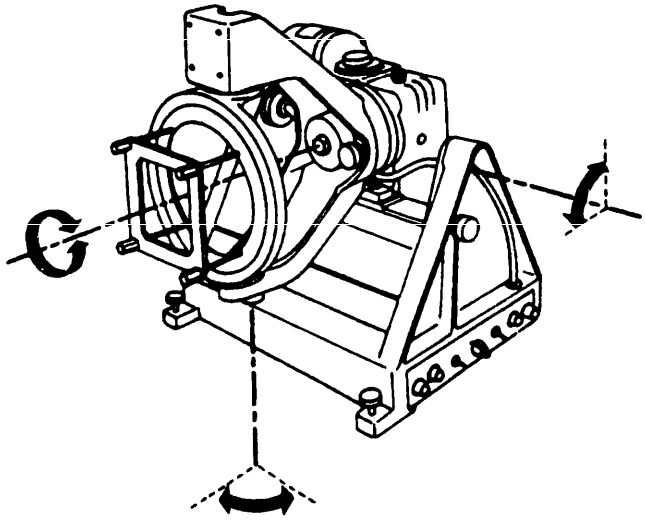


Figure 1(b). Tester with fork and cradle in roll (horizontal) position.

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